

```
In [1]: print('Addition: ', 1 + 2)
print('Subtraction: ', 2 - 1)
print('Multiplication: ', 2 * 3)
print('Division: ', 4 / 2)
print('Division: ', 6 / 2)
print('Division: ', 7 / 2)
print('Division without the remainder: ', 7 // 2)
print('Modulus: ', 3 % 2)
print('Division without the remainder: ', 7 // 3)
print('Exponential: ', 3 ** 2)
```

```
Addition: 3
Subtraction: 1
Multiplication: 6
Division: 2.0
Division: 3.0
Division: 3.5
Division without the remainder: 3
Modulus: 1
Division without the remainder: 2
Exponential: 9
```

```
In [2]: print('Floating Number,PI', 3.14)
print('Floating Number, gravity', 9.81)
```

```
Floating Number,PI 3.14
Floating Number, gravity 9.81
```

```
In [3]: print('Complex number: ', 1 + 1j)
print('Multiplying complex number: ', (1 + 1j) * (1-1j))
```

```
Complex number: (1+1j)
Multiplying complex number: (2+0j)
```

```
In [4]: a = 3
b = 2
```

```
In [7]: total = a + b
diff = a - b
product = a * b
division = a / b
remainder = a % b
floor_division = a // b
exponential = a ** b
print(total, diff, product, division, remainder, floor_division, exponential)
```

```
5 1 6 1.5 1 1 9
```

```
In [8]: print(total)
print('a + b = ', total)
print('a - b = ', diff)
print('a * b = ', product)
print('a / b = ', division)
print('a % b = ', remainder)
print('a // b = ', floor_division)
print('a ** b = ', exponential)
```

```
5
a + b = 5
a - b = 1
a * b = 6
a / b = 1.5
a % b = 1
a // b = 1
a ** b = 9
```

```
In [9]: num_one = 3
        num_two = 4
```

```
In [10]: total = num_one + num_two
        diff = num_two - num_one
        product = num_one * num_two
        div = num_two / num_two
        remainder = num_two % num_one

        print('total: ', total)
        print('difference: ', diff)
        print('product: ', product)
        print('division: ', div)
        print('remainder: ', remainder)
```

```
total: 7
difference: 1
product: 12
division: 1.0
remainder: 1
```

```
In [11]: radius = 10
        area_of_circle = 3.14 * radius ** 2
        print('Area of a circle:', area_of_circle)
```

```
Area of a circle: 314.0
```

```
In [12]: length = 10
        width = 20
        area_of_rectangle = length * width
        print('Area of rectangle:', area_of_rectangle)
```

```
Area of rectangle: 200
```

```
In [13]: mass = 75
        gravity = 9.81
        weight = mass * gravity
        print(weight, 'N')
```

```
735.75 N
```

```
In [14]: print(3 > 2)
        print(3 >= 2)
        print(3 < 2)
        print(2 < 3)
        print(2 <= 3)
        print(3 == 2)
        print(3 != 2)
        print(len('mango') == len('avocado'))
        print(len('mango') != len('avocado'))
        print(len('mango') < len('avocado'))
        print(len('milk') != len('meat'))
```

```
print(len('milk') == len('meat'))  
print(len('tomato') == len('potato'))  
print(len('python') > len('dragon'))
```

True
True
False
True
True
False
True
False
True
True
False
True
True
False

```
In [15]: print('True == True: ', True == True)  
print('True == False: ', True == False)  
print('False == False:', False == False)  
print('True and True: ', True and True)  
print('True or False:', True or False)
```

True == True: True
True == False: False
False == False: True
True and True: True
True or False: True

```
In [16]: print(3 > 2 and 4 > 3)  
print(3 > 2 and 4 < 3)  
print(3 < 2 and 4 < 3)  
print(3 > 2 or 4 > 3)  
print(3 > 2 or 4 < 3)  
print(3 < 2 or 4 < 3)  
print(not 3 > 2)  
print(not True)  
print(not False)  
print(not not True)  
print(not not False)
```

True
False
False
True
True
False
False
False
True
True
False

In []: